



Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
Integrated Bachelor's-Master's Program
End-Semester Examination-Autumn 2025

Subject: Energetics and Bonding
Full Marks: 50

Subject Code(s): CHS1101
Time Allotted: 3 hrs.

Answer any EIGHT questions

All the symbols used bear their usual meaning

All questions carry equal marks (8×6); 2 is reserved for clarity and neatness

1. Derive Gibbs' phase rule for a general multi-component, multiphase system.
2. (a) One mole of an ideal monoatomic gas is compressed isothermally from 100 L to 50 L against a constant pressure of one atmosphere. Calculate w , ΔE and ΔH .
(b) Use two laws of thermodynamics to prove $\frac{\partial P}{\partial T_V} = \frac{\partial S}{\partial V_T}$
3. (a) Prove that an isotherm cannot intersect an adiabat twice.
(b) Establish the expression for the Joule-Thomson expansion coefficient.
4. (a) Derive the Clausius-Claperon equation.
(b) Draw and illustrate a representative phase diagram to explain the triple point.
5. One mole of an ideal gas at 25°C and 1 atmosphere undergoes an isothermal expansion against zero pressure to triple its volume. Calculate the change in entropy ΔS .
6. (a) Write down the stationary Schrodinger equation for a particle moving in a one-dimensional box of length L and the boundary conditions for the wave function. Solve the equation to obtain the eigenvalues and eigenfunctions.
(b) Normalize the ground state wave function of the particle.
7. (a) Write down the Hamiltonian for a simple harmonic oscillator and express the corresponding Schrodinger equation in dimensionless form.
(b) Show that the first excited state wave function of the oscillator is of the form $X e^{-\alpha X}$, where α is a constant and X is the scaled co-ordinate. Find out α and the energy of this state.
8. (a) Explain the normal weak field Zeeman effect.
(b) Determine the eigenvalues and eigenfunctions of L_z (Z-component of angular momentum) for an hydrogenic system under appropriate boundary condition.

9. (a) Draw schematically the radial part and the contour plot of $2s$ wavefunction, $(r - 2a_0)e^{-r/2a_0}$ of hydrogen atom.

(b) Show that the eigenvalues of a Hermitian operator are real.

10. (a) Calculate the eigenvalues of $L_x^2 + L_y^2$.

(b) Show that $\frac{\partial}{\partial x}$ is not Hermitian but $\frac{\partial^2}{\partial x^2}$ is a Hermitian operator.